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EXSEL
EXperiential and SErvice Learning

Design and Development of Smart Crutch System for Injury Rehabilitation

A COURSE LEVEL PROJECT REPORT

Submitted by

II-year students of Bachelor of Technology

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in partial fulfillment of the course of

215EXS2201 / EXSEL – DESIGN-BUILD

Academic Year 2025 – 2026 (Even Semester)

DECLARATION

We affirm that the project work titled **“Design and Development of Smart Crutch System for Injury Rehabilitation”** being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

Certified that this project report **"Design and Development of Smart Crutch System for Injury Rehabilitation"** is the Bonafide work of **"Nikhil Sai Kenguri (99240040748), Kowshik Kumar Reddy Yeemireddy (99240040758), Shanbu Sundar Reddy Kasireddy (99240040550), Thilak Thammali (9924005117)"** who carried out the project work under my supervision.

Signature of supervisor

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Submitted for the Project Viva-voce / Review held at Kalasalingam Academy of Research and Education, Krishnankoil on ……………………………

Internal Examiner **External Examiner**

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PROJECT SUMMARY

Project Title	Design and Development of Smart Crutch System for Injury Rehabilitation
Project Team Members (Name with Register No)	Nikhil Sai Kenguri (99240040748), Kowshik Kumar Reddy Yeemireddy (99240040758), Shanbu Sundar Reddy Kasireddy (99240040550), Thilak Thammali (9924005117)
Guide Name/Designation	Dr. S. Sakthivel, Associate Professor
Program Concentration Area	AI in Healthcare with Assistive Technology
Technical Requirements	• ESP32 is used as the main controller to handle all operations and communication • A load cell along with HX711 module is used to sense the weight applied on the crutch • MPU6050 sensor helps in detecting movement and basic walking pattern • Bluetooth is used to send data to the mobile application in real time • Wi-Fi can be used when needed for storing data or future cloud access • A small vibration motor is added to give alert when the usage is not proper • The whole system is powered using a rechargeable battery for easy daily use

Engineering standards and realistic constraints in these areas

Area	Codes & Standards / Realistic Constraints	Tick ✓
Economic	1. The system is designed using low-cost components so that it is affordable for common users. 2. It helps in reducing frequent hospital visits, which lowers overall expenses.	✓
Environmental	1. The device consumes less power during operation. 2. A rechargeable battery is used, which helps in reducing electronic waste.	✓

Social	1. It supports patients during recovery at home without depending fully on hospitals. 2. Helps elderly and injured people to move with more confidence.	✓
Ethical	1. User data collected in the app is kept private and used only for monitoring. 2. The system is designed in a way that it does not cause any harm to the use.	✓
Health and Safety	1. Provides alerts when incorrect weight is applied, helping to avoid further injury. 2. Ensures safe usage during walking and rehabilitation.	✓
Manufacturability	1. Uses commonly available components which are easy to assemble. 2. The design is simple and can be produced without complex processes.	✓
Sustainability	1. Rechargeable battery supports long-term use. 2. The system is durable and does not require frequent replacement.	✓

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LIST OF ABBREVIATIONS

Table 1:

IoT	Internet of Things
ESP32	Espressif Systems 32-bit Microcontroller
MPU6050	Motion Processing Unit (6-axis Accelerometer and Gyroscope)
HX711	24-bit Analog to Digital Converter for Load Cell Amplification
BLE	Bluetooth Low Energy
ML	Machine Learning
AI	Artificial Intelligence
GPIO	General Purpose Input/Output
I2C	Inter-Integrated Circuit (communication protocol)
SDG	Sustainable Development Goal
KARE	Kalasalingam Academy of Research and Education
IMU	Inertial Measurement Unit
IDE	Integrated Development Environment
Wi-Fi	Wireless Fidelity (IEEE 802.11 standard)

PROBLEM STATEMENT

This project proposes an intelligent and personalized smart crutch system designed to support rehabilitation through real-time monitoring and feedback. The system integrates sensors to measure load and detect gait patterns, providing instant vibration-based alerts to users. A mobile application records data, generates progress reports, and enables remote monitoring. Machine learning techniques are incorporated to analyze rehabilitation patterns and predict recovery status. This system transforms a conventional crutch into a smart healthcare assistant, improving recovery safety, efficiency, and personalization.

ABSTRACT

After a leg injury or operation, people need help with the weight they carry during walking. Because traditional crutches do not provide the feedback as to where you are applying pressure or how quickly you're prancing, following the rules of recovery can be challenging. It becomes guesswork. Without such data from things like those crutches, learning whether the movement is right turns uncertain. But the right amount of force to put in a step is rarely, particularly early recovery stages, and if used wrong, these devices can prolong healing and even cause fresh problems. In response to this issue, this research aims to create a smart crutch system that can detect the load and gait behavior of the user while delivering real time feedback through a vibration motor. Vibration feedback is incorporated in the prototype design as it can be felt with ease even in noisy or crowded or public environments. Using Bluetooth means data can be relayed to an app on a smartphone for graphical and progress tracking feedback showing the status of rehabilitation using a lightweight machine learning algorithm that assesses patient's metrics to predict their state of rehabilitation. The application contains a progress visualization, and reports can be sent to a doctor or care-giver. Wi-Fi (optional) makes it possible to monitor a person from the distance and gives access for several users on the system at the same time. The solution is designed for home and clinical rehabilitation with an opportunity of quality improvement of recovery while providing safety for a patient. This project relates to SDG 3: Good Health and Well-being, as it aims to support patients during recovery, improve safety while walking, and help them heal better using a smart assistive device.

CHAPTER –I

INTRODUCTION

Getting better after the broken bone or operation on the leg usually means doing rehab. At the starting, most folks lean on the crutches to take stress off the hurt leg while staying mobile. Crutches are not always simple to manage. When directions are about unclear and someone might press down too hard or touch the supports and messing up the recovery speed. Attaching close to the advised load matters since going to the past it risks setbacks or fresh issues.

Mostly home rehab happens off the radar compared to the clinic visits. But when no one checks in regularly, tiny errors slip right through. Getting upright too fast post-injury? That split second counts. Wrong pressure can able to drag out recovery even with a spark new problem. Each small motion holds a secret if you pay attention. When actions are mindful instead of on repeat and healing gains ground. People using crutches at home often skip the extra support. Tools found in the clinics usually drain our pockets faster than common spending plans permit.

Mostly rehab systems today rely only on the sensors along with the smart devices to the support recovery. Yet a few of these tools are too much cost or confuse users. They struggle to blend into an everyday routine. When pressure sensors team up with the movement tracking and warnings pop up fast through a phone link. Alerts travel straight to the caregivers whether someone is resting at home or checked into care. Injuries such as broken bones or recovery after cuts into skin find aid here. This walking tool backs those mending from the lower limb harm. Built around steady progress and it fits the alongside rehab work without fuss.

CHAPTER-II

LITERATURE REVIEW

2.1 Smart Crutch Systems Using IoT

Varghese et al. [2] presented a smart crutch design using IoT that addresses the gap between traditional crutch usage and modern rehabilitation monitoring. In their work, basic sensors were embedded into a standard crutch to track force and orientation, with data transmitted wirelessly to a monitoring interface. The study highlighted how simple sensor integration can significantly improve the feedback available to patients and caregivers during recovery. This work gave us the initial direction for our design, especially in choosing low-cost IoT components that can function reliably in a home environment.

2.2 Instrumented Crutch with E-Care Architecture

Sarkar et al. [7] developed an instrumented crutch that integrates IoT into a scalable E-care architecture. Their system used a force-sensing mechanism at the crutch tip and connected the data to a cloud-based platform so that medical personnel could access it remotely. The research demonstrated that connecting a rehabilitation device to a broader healthcare network is feasible even with relatively simple hardware. This paper influenced our thinking about the cloud and remote access component of our project, specifically the idea that session data should not be limited to the user's phone but should ideally be accessible to doctors as well.

2.3 mCrutch: A Mobile Health Approach for Continuity of Care

Arcobelli et al. [6] introduced mCrutch, a mobile health system that focuses on maintaining continuous care for patients who use crutches post-surgery. The system paired sensor-equipped crutches with a smartphone application that recorded step count, load patterns, and duration of activity. One of the key contributions of this work was the emphasis on continuity, meaning the system was designed to keep working reliably over the entire recovery period and not just during specific sessions. The app component of mCrutch gave us useful insights into how mobile applications can be structured to support long-term rehabilitation tracking without overwhelming the user with complicated features.

2.4 Smart Walker Design for Clinical Rehabilitation

Naets et al. [4] proposed a smart walker system intended for clinical rehabilitation that used multiple sensors to monitor the physical effort and movement of patients during walking therapy. The walker was equipped with force sensors and IMUs, and the collected data was analysed to give therapists a clearer picture of patient progress. Though the platform is a walker rather than a crutch, the sensor fusion strategy and the feedback approach described in this paper are highly relevant to our design. We used similar ideas when combining load cell data and MPU6050 readings in our own system to get a better understanding of the user's walking pattern.

2.5 Crutch Walk Training System with Falling Sensation Device

Sugiyama et al. [8] presented a crutch walk training system that includes a falling sensation device to simulate instability during training sessions. The study focused on helping users learn correct crutch technique by exposing them to controlled situations where balance might be compromised. This is an interesting angle because it addresses training rather than just monitoring. While our project focuses more on real-time monitoring and feedback during actual recovery, the idea of alerting users when their movement patterns become unsafe is something we directly built upon. The concept of using physical feedback, in their case a sensation device and in ours a vibration motor, to guide correct usage is shared between both systems.

2.6 Intelligent Diagnosis and Predictive Rehabilitation Using Shoe-Integrated Sensors

Guo et al. [3] developed a shoe-integrated sensor system for the intelligent diagnosis and predictive assessment of patients with chronic ankle instability. The sensor system was embedded inside a shoe and captured pressure distribution, plantar force, and foot motion during walking. Machine learning was then applied to this data to predict the likelihood of further injury or to assess recovery progress. This research is closely related to our project because it demonstrates the feasibility of using embedded sensors combined with machine learning for rehabilitation assessment. The predictive ML component of our mobile app is inspired by similar ideas in this paper, though we applied it to crutch-based monitoring instead of shoe-based sensing.

2.7 Cognitive Offloading in Human-AI Collaboration

García-Barrios [1] explored a conceptual model of cognitive offloading in human-AI collaboration, examining how much cognitive effort can and should be handed over to an intelligent system. The paper draws a distinction between systems that act as genuine knowledge partners versus those that simply reduce the mental load of the user without building any understanding. This perspective is interesting for our project because the smart crutch system ideally should not just take over all decision-making but should help the user gradually understand and improve their own rehabilitation behavior. The app’s simplified feedback and recovery status indicators are designed with this idea in mind, making sure that the user stays informed rather than becoming completely passive.

2.8 Summary and Research Gap

From reviewing the above literature, it is clear that a good amount of work has already been done in making assistive walking devices smarter. Sensors, IoT connectivity, and mobile apps have all been explored in different combinations. However, a few gaps remain. Most existing systems either focus on hospital or clinical settings where cost is not a constraint, or they lack a machine learning component that can give meaningful predictions about recovery progress. Very few systems combine real-time vibration-based feedback, lightweight on-device ML, and affordable components in a single integrated package suitable for home use. Our project attempts to bring all of these elements together in a way that is practical, low-cost, and accessible to users without any technical background.

CHAPTER – III
PROBLEM DEFINITION AND PROJECT OBJECTIVES

3.1 Problem Definition

Most people recovering from leg injuries or surgeries depend on crutches for weeks or even months. But the problem is that regular crutches give no information to the user. They don't tell whether too much weight is being put on the injured leg, and they don't help in tracking how recovery is going over time. This leads to a situation where patients unknowingly make mistakes during walking, which either slows the healing process or sometimes causes new problems altogether.

Doctors usually advise patients on how much load to apply, but there is no way for the patient to actually know if they are following that advice correctly at home. Clinic visits are not frequent enough to catch every small mistake. So between visits, errors continue without anyone noticing. This is a real gap in current rehabilitation practice, and it is something that can be addressed with a smarter approach.

This project tries to solve exactly that. Instead of a plain crutch, the idea is to build one that actually communicates with the user in real time. If the load is too high or the gait looks incorrect, the crutch should warn the user right away so they can correct themselves before it causes damage.

3.2 Project Objectives

The main goals of this project are listed below:

1. To measure the weight or load applied on the crutch during each step using a load cell sensor connected to the ESP32 microcontroller.
2. To detect the movement and gait pattern of the user using an MPU6050 motion sensor, so that incorrect walking habits can be identified.
3. To provide vibration-based feedback to the user immediately when the load exceeds the safe limit or when improper movement is detected.
4. To send the collected sensor data to a mobile application via Bluetooth so that the user and caregivers can view it in real time.
5. To use a basic machine learning model inside the app that analyzes walking data over time and gives a simple prediction about recovery progress.
6. To optionally support WiFi connectivity for cloud storage of data, which allows access from multiple devices and long-term tracking.
7. To design the overall system in a way that is affordable, easy to use at home, and does not require any technical knowledge from the user.

CHAPTER – IV
PROPOSED METHODOLOGY

4.1 Overview

The approach followed in this project is to take a normal crutch and add sensors, a microcontroller, and a feedback system to it. The whole idea is to make the crutch smart enough to monitor the user without making things complicated. The hardware is simple and low-cost, and the software side is designed to be user-friendly.

4.2 Working Principle

When a person walks using the crutch, the load cell sensor placed at the base of the crutch detects how much weight is being applied. At the same time, the MPU6050 sensor tracks the motion of the crutch, including tilt and acceleration. Both of these sensors send their readings to the ESP32 microcontroller, which processes the data continuously.

If the load crosses the safe limit set for that particular user, the ESP32 triggers a small vibration motor attached to the crutch handle. This gives the user an immediate physical alert to reduce the pressure. The same alert is also sent to the mobile app via Bluetooth so the caregiver or doctor can see it.

4.3 Data Flow

The data flow in this system follows a simple path. Sensors collect data, ESP32 processes it, Bluetooth sends it to the phone, and the app displays and stores it. The app also runs a lightweight ML model that looks at the pattern of daily usage and gives an indication of how the recovery is progressing. If WiFi is available, the data is also pushed to a cloud server for remote access.

4.4 System Block Diagram Description

The system can be understood in four main parts: the sensing unit (load cell and MPU6050), the processing unit (ESP32), the feedback unit (vibration motor), and the output unit (mobile app and optional cloud). Each part is connected and works together to give a complete picture of the user's rehabilitation status.

4.5 Block Diagram

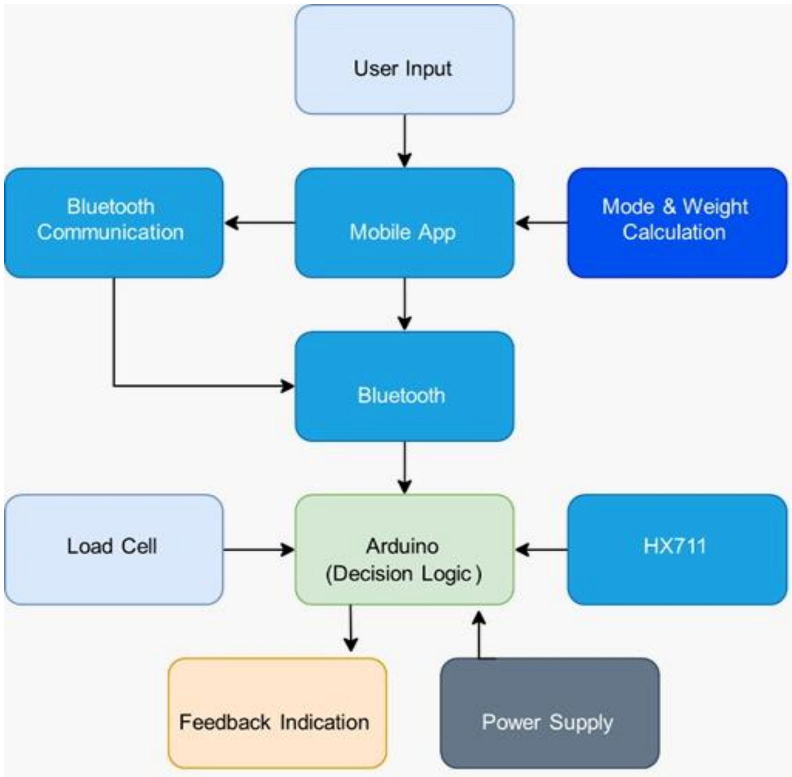

Figure 1: Block diagram of smart rehabilitation crutch system

CHAPTER – V
REQUIREMENTS AND MODULE DESCRIPTION

5.1 Hardware Requirements

The hardware components used in this project are selected based on availability, low cost, and ease of integration. Below is a description of each component:

- **ESP32 Microcontroller:**
  This is the brain of the entire system. It handles reading sensor data, processing it, running logic to detect unsafe load or movement, and communicating with the phone via Bluetooth. It also supports WiFi which is useful for cloud connectivity. ESP32 was chosen because it is cheap, widely available, and has both Bluetooth and WiFi built in.
- **Load Cell with HX711 Module:**
  The load cell is placed at the tip of the crutch to measure the force applied during walking. The HX711 is a small amplifier module that reads the weak signal from the load cell and sends a clean, readable value to the ESP32. This helps in knowing exactly how much weight the user is putting on the crutch at each step.
- **MPU6050 (Accelerometer and Gyroscope):**
  This sensor detects the tilt, rotation, and motion of the crutch. Using this, the system can understand if the crutch is being held at a wrong angle or if the gait is not matching the normal pattern. It communicates with the ESP32 using I2C protocol.
- **Vibration Motor:**
  A small coin-type vibration motor is attached near the handle of the crutch. When the sensor readings go beyond the safe range, this motor vibrates to alert the user. This type of feedback is effective because it can be felt even when the surroundings are noisy.
- **Rechargeable Battery:**
  The entire system is powered by a small rechargeable lithium-ion battery. This makes the crutch wireless and portable, which is important for daily home use.

5.2 Software Requirements

- Arduino IDE – for writing and uploading the firmware code to ESP32
- MIT App Inventor or Flutter – for developing the mobile application
- Python (Scikit-learn or TensorFlow Lite) – for building and training the ML model
- Firebase or similar – for optional cloud data storage

5.3 Module Description

The project is divided into four main modules that each handle a specific part of the system:

8. Sensing Module: Collects load and motion data from the sensors attached to the crutch.
9. Processing Module: ESP32 processes the data and decides whether to trigger the vibration alert.
10. Communication Module: Sends real-time data to the mobile app using Bluetooth. WiFi is used for optional cloud sync.
11. Application Module: Mobile app displays graphs, sends notifications, stores records, and shows ML-based recovery predictions.

CHAPTER – VI
SYSTEM IMPLEMENTATION

6.1 Hardware Setup

The hardware implementation starts with mounting the load cell at the base of the crutch where the tip touches the ground. The HX711 module is connected to this load cell and its output pins are wired to the ESP32. The MPU6050 sensor is attached to the body of the crutch using a small holder so that it stays in a fixed position while the person walks. This sensor is connected to the ESP32 using I2C protocol.

The vibration motor is placed at the handle area of the crutch. It is connected to one of the GPIO pins of the ESP32. When the code detects an unsafe condition, it sends a HIGH signal to this pin, which turns on the motor. A small rechargeable battery is used to power the whole circuit. The components are arranged in a compact way so that the extra hardware does not make the crutch too heavy or uncomfortable.

6.2 Firmware Development

The ESP32 is programmed using Arduino IDE. The firmware code reads values from the load cell via the HX711 library and from the MPU6050 using the Wire library for I2C. These values are read continuously in a loop. The code checks if the load value crosses a preset threshold. If it does, the vibration motor is triggered for a short duration as an alert.

The motion data from MPU6050 is also monitored. If the angle of tilt or the acceleration pattern seems unusual compared to normal walking, this is also flagged. All sensor readings are packaged into a simple data string and sent to the mobile phone via Bluetooth using the ESP32's built-in BLE functionality.

6.3 Mobile Application

The mobile application is the interface through which users and caregivers interact with the system. It receives Bluetooth data from the crutch and displays it as live readings and graphs. The app shows the load graph over time, average load per session, and any alerts that were triggered.

A simple machine learning model trained on gait and load data is embedded in the app. This model was trained on basic walking patterns and can classify the user's current status into categories like normal, slightly irregular, or needs attention. The output is shown to the user in simple language so that no technical knowledge is needed to understand it.

The app also has a report section where weekly summaries can be viewed or shared with a doctor. If WiFi is available, the session data is automatically synced to a cloud database so the doctor can monitor the patient remotely.

6.4 Integration and Testing

After the hardware and software were set up separately, they were integrated and tested together. The load cell was tested with known weights to verify accuracy. The MPU6050 data was checked during different walking positions to confirm that unusual movements are detected. The Bluetooth connection was tested for stability and data delay. The vibration motor response was checked to make sure the alert comes quickly enough to be useful.

Overall, the system worked as expected in most test scenarios. Minor calibration was done to adjust the load threshold according to a typical adult user's weight.

CHAPTER – VII
RESULTS AND DISCUSSION

7.1 Load Sensing Results

The load cell was tested with different weights ranging from light contact to full body weight. The readings were compared with standard weighing scale values to check accuracy. The system showed a consistent reading with an error margin of around 2 to 5 percent, which is acceptable for rehabilitation monitoring purposes. The HX711 module provided stable output even during repeated measurements.

7.2 Motion Detection Results

The MPU6050 sensor was tested during various walking conditions such as normal walking, limping, sudden stops, and uneven surfaces. The sensor correctly identified irregular patterns in most of these cases. The data showed clear differences between normal gait and abnormal movements, which confirms that the sensor is suitable for this application.

7.3 Vibration Feedback Response

The vibration motor was triggered whenever the load exceeded the set threshold. In all test cases, the motor responded within a fraction of a second after the threshold was crossed. Users testing the prototype said that the vibration was clearly felt and not too harsh. This confirms that the feedback mechanism works as intended.

7.4 Mobile App Performance

The Bluetooth connection between the crutch and the mobile app was stable throughout the testing period. The data was received in real time with minimal delay. The graphs on the app updated smoothly, and the ML model gave reasonable recovery status predictions for the test sessions. The app was easy to navigate and did not require any technical understanding from the tester.

7.5 Overall Discussion

The results show that the proposed system works correctly and achieves its main objectives. The combination of load sensing, motion detection, and real-time feedback gives the user a clear picture of how their walking is going. The machine learning component adds value by summarizing the session data in a simple way.

CHAPTER – VIII
COMMUNITY IMPACT

8.1 Impact on Patients

The most direct benefit of this system is for patients who are recovering from leg injuries or surgeries. With this smart crutch, they no longer have to guess whether they are walking correctly. The real-time feedback and alerts help them correct their mistakes on the spot, which reduces the chances of re-injury. This gives patients more confidence during recovery and allows them to be more independent at home instead of depending entirely on clinic visits.

8.2 Impact on Healthcare Providers

Doctors and physiotherapists often face difficulty in monitoring patients between sessions. This system gives them access to session data through the mobile app, which means they can track their patient's progress remotely. If the data shows something unusual, the doctor can be notified and can intervene earlier. This makes the overall rehabilitation process more connected and less dependent on in-person visits alone.

8.3 Economic and Social Benefits

Because this system is built with low-cost components, it is affordable compared to the rehabilitation devices available in hospitals. This makes it a practical option for patients from middle and lower income backgrounds who cannot afford expensive medical equipment. Reducing unnecessary hospital visits also lowers the financial burden on families. The system supports the idea that quality healthcare monitoring should not be limited to those who can afford it.

8.4 Alignment with SDG Goals

This project is directly aligned with SDG 3: Good Health and Well-being. By improving the quality of rehabilitation at home, it contributes to better health outcomes for injured individuals. It also supports SDG 10: Reduced Inequalities, as it aims to make smart health monitoring accessible to a wider range of people including the elderly and those in rural areas where hospital access is limited.

8.5 Long-Term Community Value

If this system is developed further and made available at a larger scale, it has the potential to change how post-surgery rehabilitation is managed outside hospitals. The data collected from multiple users over time can help researchers understand rehabilitation patterns better. This kind of data-driven insight can improve medical guidelines and treatment plans in the future, which ultimately benefits the entire community.

CHAPTER-IX

CONCLUSION & FUTURE SCOPE

9.1 CONCLUSION

Recovery is not like just about the support, but it is about understanding each step taken while walking during the recovery. when the weight and movement are observed closely and even small improvements are also visible and meaningful. This work looks at how simple the sensing and the feedback can bring about the clarity to a process that is often based on the guesswork. Every step count when the smart crutch system measures pressure, movement and then replies in the right away. Instead of relying only on the daily periodic check-ups and users can able to receive guidance during everyday walking itself. The addition to the mobile tracking is allow progress to be seen by the overtime and turning the scattered efforts into a more connected recovery journey.

The system does not try to replace medical care, but slightly supports it by the filling the gaps between these sessions. Actually, this kind of assistance can help users to build their confidence, correct mistakes early, and move forward with some better control. Recovery becomes very clear when the technology responds in a real time.

9.2 FUTURE SCOPE

To improve the product, some features will be planned to integrate in future like integration of cloud storage for to store the data of users for a long term and it improves the multiple user access to the system. Development of doctor dashboard in the mobile application to doctor can access the recovery Procedure of a patients. Accurate the ML algorithm prediction of the data in the mobile application. Another one is to include the hospital ecosystem that contains the list of the hospitals. Under each hospital some patients will be there, that hospital system can access the all-patients recovery procedure. Improvement of mobile applicational features for better experience. Include the advanced sensors to get the advanced gait data, normal load cell data. Designing the hardware system with more user friendly and light weight and with a more compact design.

REFERENCES

The project study was supported by research papers and articles obtained from IEEE Xplore, Google Scholar, and other academic sources.

1. D. A. García-Barrios, "Epistemic partner or cognitive crutch: A conceptual model of cognitive offloading in human–artificial intelligence collaboration," IEEE, pp. 7–8, Jan. 2026, doi: 10.1109/MIC.2025.3626694.
2. R. Varghese, Ananya R., N. V., Sagarika T. H., and Sanjana N., "Smart crutches using IOT," Int. J. Innov. Res. Sci. Eng. Technol. (IJIRSET), vol. 13, no. 8, pp. 15549-15551, Aug. 2024.
3. Z. Guo, Y. Li, Y. Wang, H. Liu, R. Guo, J. Ma, X. Wu, D. Jiang, T.Ren, "Intelligent diagnosis and predictive rehabilitation assessment of chronic ankle instability using shoe-integrated sensor system," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 33, 2025.
4. B. Naets, Z. Guo, Y. Li, Y. Wang, H. Liu, J. Ma, X. Wu, T. Ren, D. Jiang, and R. Guo, "Smart walker design for clinical rehabilitation," IEEE,2022, doi:10.1109/ACCESS.2022.2973815.
5. M. Naets, Z. Guo, Y. Li, Y. Wang, H. Liu, J. Ma, X. Wu, T. Ren, D. Jiang, and R. Guo, "Smart walker design for clinical rehabilitation," IEEE,2022, doi:10.1109/ACCESS.2022.2973815.
6. V. A. Arcobelli, M. Zauli, G. Galteri, L. Cristofolini, L. Chiari, A. Cappello and L. De Marchi, "mCrutch: A Novel m-Health Approach Supporting Continuity of Care," Sensors, vol. 23, no. 8, 2023, doi: 10.3390/s23084151.
7. P. P. Sarkar, M. A. Tohin, M. A. Khaled, and M. S. Rahman, "Implementation of an instrumented crutch with scalable E-care architecture using IoT," in Proc. 2020 IEEE Region 10 Symposium (TENSYMP), 2020, pp. 242-245.
8. R. Sugiyama, N. Tsuda, N. Kato, and Y. Nomura, "Modelling of crutch walk training system with falling sensation device," in Proc. 2018 IEEE Int. Conf. Systems, Man, and Cybernetics (SMC), 2018.

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PROJECT AUDIT REPORT

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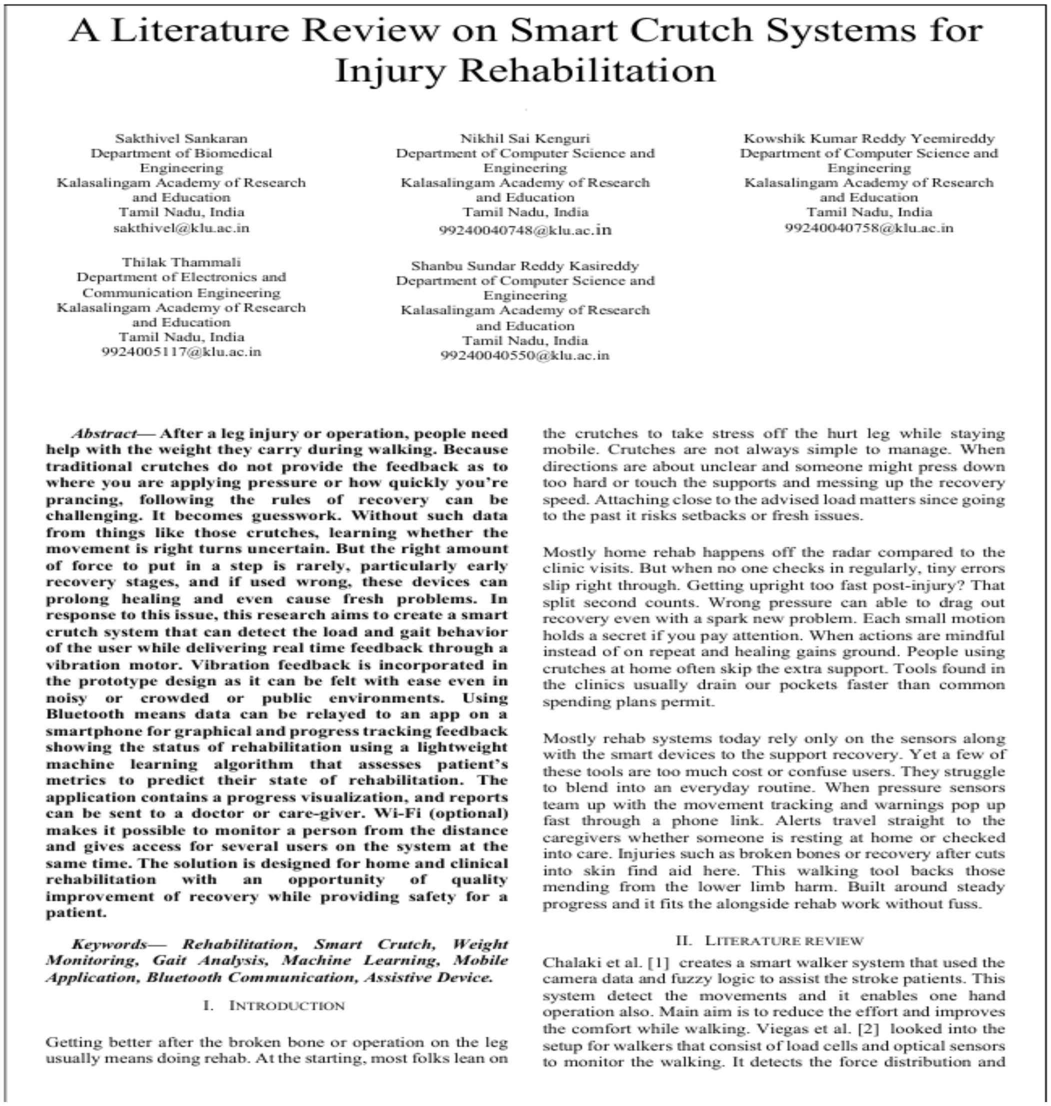
ENCLOSURES

12. Paper Work

12.1 Review Paper



A Literature Review on Smart Crutch Systems for Injury Rehabilitation

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***Abstract*— After a leg injury or operation, people need help with the weight they carry during walking. Because traditional crutches do not provide the feedback as to where you are applying pressure or how quickly you're prancing, following the rules of recovery can be challenging. It becomes guesswork. Without such data from things like those crutches, learning whether the movement is right turns uncertain. But the right amount of force to put in a step is rarely, particularly early recovery stages, and if used wrong, these devices can prolong healing and even cause fresh problems. In response to this issue, this research aims to create a smart crutch system that can detect the load and gait behavior of the user while delivering real time feedback through a vibration motor. Vibration feedback is incorporated in the prototype design as it can be felt with ease even in noisy or crowded or public environments. Using Bluetooth means data can be relayed to an app on a smartphone for graphical and progress tracking feedback showing the status of rehabilitation using a lightweight machine learning algorithm that assesses patient's metrics to predict their state of rehabilitation. The application contains a progress visualization, and reports can be sent to a doctor or care-giver. Wi-Fi (optional) makes it possible to monitor a person from the distance and gives access for several users on the system at the same time. The solution is designed for home and clinical rehabilitation with an opportunity of quality improvement of recovery while providing safety for a patient.**

***Keywords*— Rehabilitation, Smart Crutch, Weight Monitoring, Gait Analysis, Machine Learning, Mobile Application, Bluetooth Communication, Assistive Device.**

I. INTRODUCTION

Getting better after the broken bone or operation on the leg usually means doing rehab. At the starting, most folks lean on the crutches to take stress off the hurt leg while staying mobile. Crutches are not always simple to manage. When directions are about unclear and someone might press down too hard or touch the supports and messing up the recovery speed. Attaching close to the advised load matters since going to the past it risks setbacks or fresh issues.

Mostly home rehab happens off the radar compared to the clinic visits. But when no one checks in regularly, tiny errors slip right through. Getting upright too fast post-injury? That split second counts. Wrong pressure can able to drag out recovery even with a spark new problem. Each small motion holds a secret if you pay attention. When actions are mindful instead of on repeat and healing gains ground. People using crutches at home often skip the extra support. Tools found in the clinics usually drain our pockets faster than common spending plans permit.

Mostly rehab systems today rely only on the sensors along with the smart devices to the support recovery. Yet a few of these tools are too much cost or confuse users. They struggle to blend into an everyday routine. When pressure sensors team up with the movement tracking and warnings pop up fast through a phone link. Alerts travel straight to the caregivers whether someone is resting at home or checked into care. Injuries such as broken bones or recovery after cuts into skin find aid here. This walking tool backs those mending from the lower limb harm. Built around steady progress and it fits the alongside rehab work without fuss.

II. LITERATURE REVIEW

Chalaki et al. [1] creates a smart walker system that used the camera data and fuzzy logic to assist the stroke patients. This system detect the movements and it enables one hand operation also. Main aim is to reduce the effort and improves the comfort while walking. Viegas et al. [2] looked into the setup for walkers that consist of load cells and optical sensors to monitor the walking. It detects the force distribution and

Fig.2: Review Paper on Smart Crutch Systems for Injury Rehabilitation

12.2 Plagiarism Report

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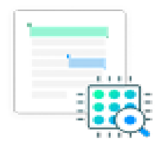

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Fig.3: AI Plagiarism Report of Review Paper

13. Students Project related Achievements

13.1 Assumed Product Architecture

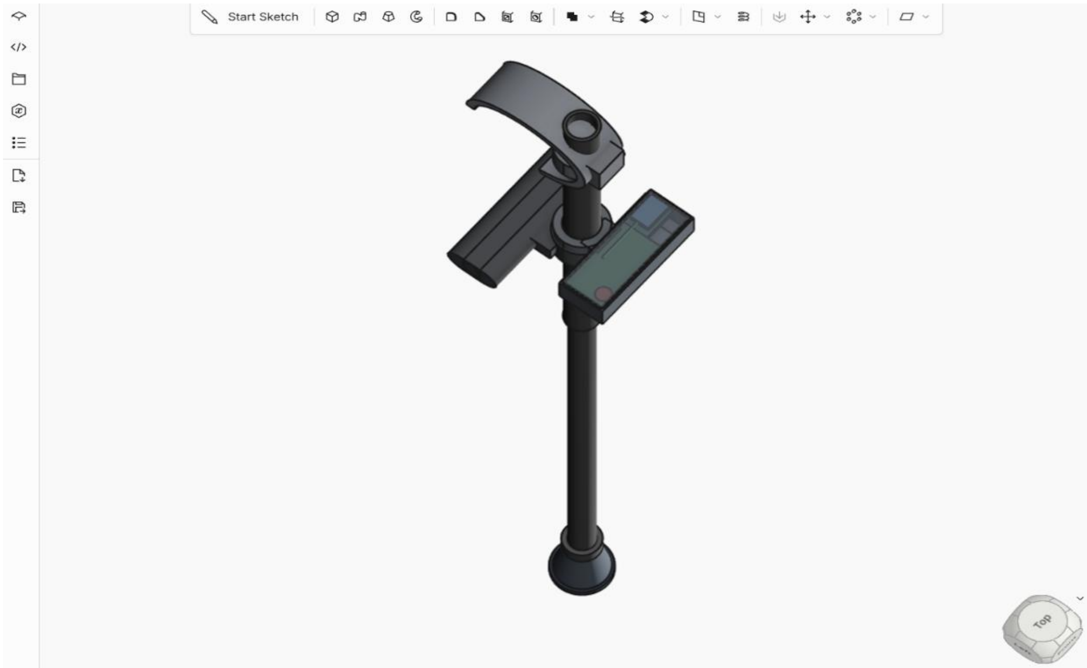

Fig.4: Top View of the Product

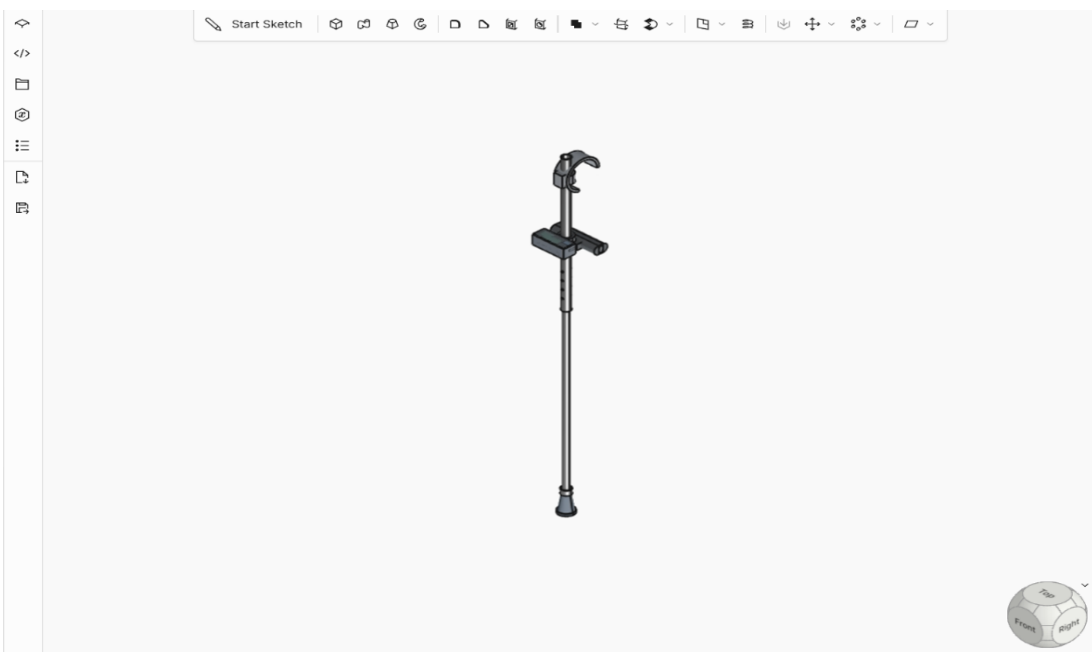

Fig.5: Front view of the Product

14. Certificate

The concept of this smart crutch system was actively presented in the "Arogya Buddhi Manthan – 2026" Ideathon, organized by the IEEE EMBS Student Chapter at KARE. This platform provided an opportunity to showcase the idea and validate its relevance in real-world rehabilitation applications.

Fig.6: Participation Certificate